

Explicit quantification of coastal cultural ecosystem services: A novel approach based on the content and sentimental analysis of social media

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ABSTRACT

Ecosystem service assessments have been conducted for coastal management with a primary focus on tangible natural capital. Thus far, the explicit variation of cultural ecosystem services (CESs) and particularly the sentiments of CES-related ecotourism are not well understood. This paper takes advantage of big social media data to unravel the patterns of CESs and visiting sentiments in coastal areas of Hong Kong. Through machine-learned keyword labels, we employed content analysis to derive visual information for geotagged photographs. Applying natural language processing to apps with machine learning in cloud computing, we derived the sentiments based on user-generated textual content associated with coastal ecotourism. Based on regression analysis and multiple comparisons analysis, we identify the association between critical demographic and temporal factors with CES and related visiting sentiments. Referring to previous studies, we identified the main intangible benefits into six basic divisions based on 424 keyword labels for coastal areas. Our results show that hotspots of CESs are spatially concentrated in both cultural attractions and protected areas, which are critical for coastal ecosystem management and protection. More specifically, these areas of high CES value have good spatiotemporal accessibility, high infrastructure coverage, and spatially explicit population and economic growth. Furthermore, we discover that sentiments related to coastal CESs vary based on social media characteristics. Our study renews the indicators of quantitative CES evaluation based on crowdsourcing geospatial data.

1. Introduction

Areas within 60 miles of a coastline, or coastal areas as they are called in the literature, accommodate approximately 40% of the world's population (MEA, 2005; Martin et al., 2016) and play an essential role in improving human physical and mental health and enhancing social cohesion among communities. However, intensifying anthropologic activities within coastal areas have greatly impaired coastal and marine ecosystems and posed threats to societal sustainability. For example, the rapid urbanization in coastal cities heavily consumes the vegetation and further accelerates the coastal erosion process, which increases the vulnerability to hazards (Su et al., 2015). In response, both the scientific and political communities worldwide have called for efficient countermeasures, and an ecosystem service approach has gradually received support. In particular, the ecosystem service approach outlines the recognition and quantification of how much benefit humans obtain from natural resources (Kumar, 2010; Costanza et al., 2017), laying a basis for evaluating ecological security and formulating ecological compensation

measures for coastal areas. However, public perception of coastal ecosystems is generally underrepresented in ecosystem service assessments, and most studies have focused on qualitative insights of ecotourism visiting sentiments (Toivonen et al., 2019; Teles da Mota and Pickering, 2020).

The essential component of ecosystem service flows from coastal ecosystems to human beneficiaries is mainly nonmaterial, and very often these intangible dimensions matter more to people than material benefits (Kobryn et al., 2018). Including cultural dimensions in coastal management will address further concerns of those most affected by environmental and institutional changes (Dam Lam et al., 2019). Among the diverse ecosystem services provided in coastal areas, cultural ecosystem services (CESs) have been prioritized for the manifold contributions to human wellbeing, ecosystem integrity, and viable economies (Poe et al., 2014). CESs can comprehensively enrich the human and spiritual sides of life; encourage cognitive development; and provide opportunities for reflection, recreation, and aesthetic experiences (Drius et al., 2019; Havinga et al., 2020). One popular approach to quantifying

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CESs is that of nonmarket valuation, which determines people's willingness to maintain or protect coastal ecosystems using the travel cost method, consumer surplus (Pouso et al., 2018), total economic value analysis, and the hedonic pricing method (Sangha et al., 2019). However, this kind of method has been frequently criticized for exhibiting significant uncertainty and subjectivity in selecting samples and for showing limitations in identifying the inherent complexities and spatiotemporal dynamics of CESs. Additionally, the labor and time costs of conducting interviews are extremely high, largely reducing the possibility of large-scale projects.

Currently, the increasing popularity of social media outlets such as Instagram, Twitter, and Flickr have generated an inexhaustible supply of big data, creating new means of tackling the abovementioned challenges. In particular, we can successfully extract rich semantic and sentimental information from geotagged photographs and text posted on social media (Runge et al., 2020; Song et al., 2020) and further use this information to understand how people value and perceive CESs. A shared vision has formed such that social media-based analysis has become a new and efficient approach for the quantification of CSE. For example, data from Flickr were used to assess spatial patterns of CESs in different landscapes based on where photos were taken (Balzan and Debono, 2018; Sinclair et al., 2019). Several attempts have been made to analyze the content of images to gauge how people value natural environments (Lee et al., 2019; Song et al., 2020; Teles da Mota and Pickering, 2020) and even interactions with the natural world across a continent (Van Zanten et al., 2016). However, recent studies still show limitations. First, research on this subject has been mostly restricted to a focus on limited landscapes of CESs, while coastal areas are underrepresented, especially on the spatial scale of sustainable urban land-use planning and management (Smart et al., 2021). Second, although many studies have found that proximity to coastal areas leads to high CES values (Van Zanten et al., 2016), existing research does not fully explain which biophysical or social and infrastructure factors affect CES in coastal areas. Third, thus far, there has been little systematic research on how people's perceptions of and feelings toward CESs infer whether people will participate in coastal protection and the likelihood of engaging in behaviors supporting the desired CES. In contrast, sentiment analyses on open-ended text are becoming an important component in understanding public satisfaction with participatory GIS (Liu et al., 2020; Teles da Mota and Pickering, 2020).

Our specific objectives are to (1) identify important CESs of coastal areas through content analysis of social media images and reveal the spatial distribution of CESs; (2) combine sentimental analysis of CES-related photo metadata (user-generated textual content) into the coastal CESs assessments; (3) quantify the relationship between CESs and related sentiments with temporal and sociodemographic indicators; and (4) provide theoretical guidance for coastal ecosystem management and protection.

2. Materials and methods

2.1. Methodological flow

It takes four main steps to complete the entire analysis in our study (Fig. 1). Firstly, we use a web-crawler tool to garner image data on Flickr and related textual data. Secondly, CESs are identified through the content analysis of images. The results returned as keyword labels reflect the dominant entities of images. Thirdly, CES-related sentiments are quantified by machine learning in cloud computing through natural language possessing of user-generated text. Fourthly, potential determinants of CESs are selected and regression analysis is established to unravel the relationship between social media characteristics and CESs as well as related sentiments.

2.2. Study area

Hong Kong is located in southern China (Fig. 2) and has a coastline covering over 80,000 ha (Marafa, 2008). It has always been an important trading and tourism gateway city in Asia and worldwide. As "blue ecotourism" has been identified as a key sector of the tourism industry (Phelan et al., 2020), the Hong Kong government has been working on giving full play to its advantages and tapping its potential to create an attractive city waterfront that provides quality public spaces (Cheung and Jim, 2013). Moreover, Hong Kong has applied a coastal ecological conservation policy that maintains the sustainable development of CESs and fosters strong public awareness of environmental protection.

Human activity intensity (i.e., aquaculture, fishing, foreign trade, shipping port transportation, and leisure tourism) has continued to increase rapidly in the recent past (He et al., 2021), which raises concerns about potential threats to the sustainability of coastal areas. For example, the expansion of construction land has exceeded the growth rate of renewable biocapacity, posing a threat to natural habitats with important ecological value in coastal regions (i.e., mangrove wetlands, coral reef wetlands, and seagrass beds) (Bi et al., 2020). Consequently, in-depth information about the current and potential impacts on coastal ecosystem services is urgently needed. Hong Kong mirrors coastal areas sustaining high pressures from human society across the globe, thereby providing a typical study case for the spatial quantification of CESs in coastal areas.

2.3. Definition and classification of CESs

For defining CES, our classification process was compatible with the most representative conceptual thinking in the literature. The CESs subjected to this analysis were defined based on types in three leading ecosystem classification frameworks, including Millennium Ecosystem Assessment (MEA, 2005), The Economics of Ecosystems and Biodiversity

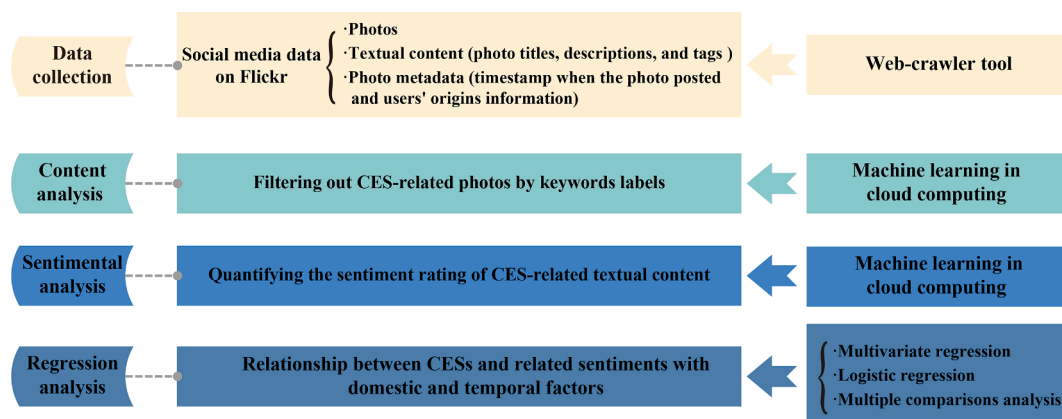


Fig. 1. Methodological flow for the quantification of cultural ecosystem services (CES) in coastal areas through social media data.

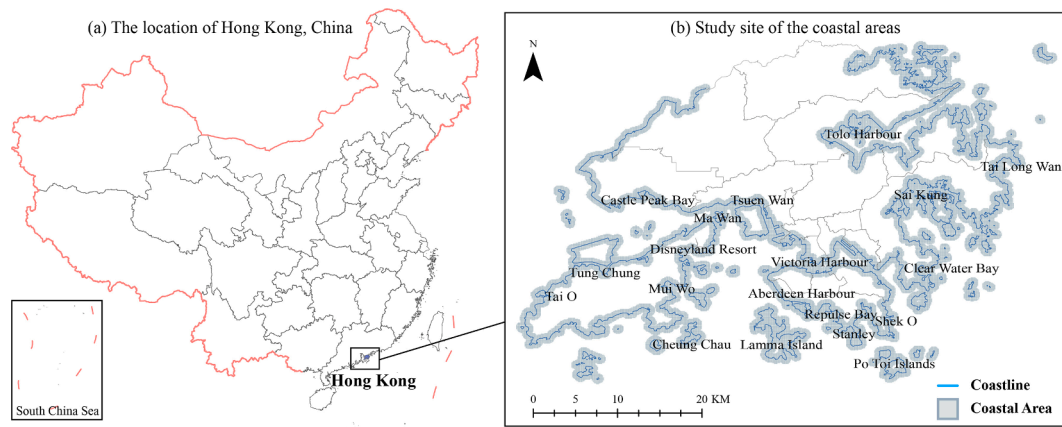


Fig. 2. Study site of the coastal areas involved in the study of Hong Kong, China.

(TEEB) (Kumar, 2010), and Common International Classification of Ecosystem Services (CICES) (Haines-Young and Potschin-Young, 2018) classification frameworks, of which physical, intellectual, spiritual values of the natural environment were included. We proposed six general service divisions: recreational, aesthetic, amenity, knowledge, naturalistic, and spiritual. Particularly, to reflect the generalized interests and concerns of government and residents of Hong Kong and other coastal areas, the service contributed to urban recreations were emphasized (Liquete et al., 2013; Liquete et al., 2016). Our classification process also focused on these services which can describe very relevant natural or socio-economic characteristics in coastal areas (Alieva et al., 2021; de Juan et al., 2021).

- (1) Recreational services. Recreational services represent the cultural benefits arising from sports-related activities, which include land and water recreation (Jackson et al., 2013; Balzan and Debono, 2018; Havinga et al., 2020). Many outdoor recreational activities in coastal areas are dependent on the availability and distribution of ecosystems, including those for jogging, cycling, and sailing. Recreational services benefit people's health and well-being by providing a pleasant physical environment.
- (2) Aesthetic services. Aesthetic services are the combination of the pluralistic attractiveness of nature from the perspective of different civilizations. Coastal ecosystems are composed of various landscape elements, and their appreciation is evaluated from specific landscape elements and overall preferences. Landscape appreciation refers to admiration primarily of vast and open landscapes such as seascapes (Ruiz-Frau et al., 2020). Meanwhile, a preference for natural structures refers to landforms associated with recreation and aesthetics, including rivers, mountains, beaches, and so on (Brown and Brabyn, 2012).
- (3) Amenity services. Amenity services represent the impact of natural components on human communities (Cochrane, 2006). A comfortable environment is visible and unique for both visitors and residents with reliable infrastructures in coastal attractions (Carreiro et al., 2008). With social development, people pay more attention to leisure activities. Beautiful public spaces from street-level amenities to urban residence structures are a prerequisite for integration with natural and cultural ecosystem services (Su et al., 2021).
- (4) Knowledge services. From indigenous knowledge systems to modern science, knowledge of the ecological environment is essential for the sustainable use of natural resources, climate change adaptation, and ecosystem resilience (Díaz et al., 2018; Haines-Young and Potschin-Young, 2018). The coastal ecosystem provides a material basis for scientific research and education. For example, coastal and oceanic observatories offer services for

delivering high-quality environmental data to meet societal and political needs (El Serafy, 2020). Such an ecosystem enables people to draw on relevant knowledge identified as knowledge services. Recently, social media propagation has raised public awareness of the natural environment witnessed by both celebrities and ordinary people (Richards and Friess, 2015).

- (5) Naturalistic services. Proenvironmental attitudes and behaviors and consciousness of mediating a proper relationship between all living organisms on Earth (Katz-Gerro and Orenstein, 2015) inspire people to pursue a greater sense of spiritual well-being. Notably, the coastal biome is one of the habitats with the greatest diversity of species in the world (Wolanski and McLusky, 2011). However, unique biomes such as mangroves and wetlands near coastal cities are threatened by extensive reclamation projects, depleting fish stocks, and coastal waters tainted by domestic waste and pollutants (Friess, 2017; Garcia Rodrigues et al., 2017). Thus, an assessment of naturalistic services is a preliminary practice for developing sustainable coastal tourism.
- (6) Spiritual services. A sacred landscape in the wild is central to religious tourism or secular pilgrimage, arousing beliefs of the past and promoting a heritage of cultural memory (Hernández-Morcillo et al., 2013; Clemente et al., 2019). In residing in a coastal environment for centuries, communities with racial and ethnic differences have particular cultures of entertainment, recreation, social customs, and habits that shape religious beliefs and essential elements of their lives.

2.4. Measuring coastal CESs and related sentiments

2.4.1. Data collection

Image data were collected from a photo-sharing website Flickr.com. Flickr is a popular source for ecotourism and recreation research (Schirpke et al., 2021). Researchers have used photos posted on Flickr at a range of spatial scales to study and found that Flickr data are reliable to analyze the geographical preferences of international tourists in China (Su et al., 2016). These publicly accessible photos on Flickr not only contain a wealth of geographic information, but also convey people's behavioral views of public green spaces such as urban parks (Song et al., 2020), conservation areas (Runge et al., 2020), and water bodies (Sinclair et al., 2018; Schirpke et al., 2021). In contrast, it is more difficult to access geolocation metadata from other social media platforms (i.e., Twitter, Instagram, and Weibo) than Flickr. These platforms are increasingly limiting the provision of accurate geolocation data due to privacy concerns (Teles da Mota and Pickering, 2021).

We accessed Flickr through its Application Programming Interface (<https://www.flickr.com/services/api/>) with a Python package flickrapi (<https://stuvel.eu/software/flickrapi/>) to download geotagged

public photos and associated metadata. The data include photo ID, user ID, coordinate, photo title, description, tag, and date data and information about the user's location. Subsequently, from the collected photos, those with missing attributes such as (1) hometown or current locations and (2) textual content including photo titles, descriptions, and tags were removed. The results are presented in aggregated form to ensure privacy.

2.4.2. Photographs selection

In the literature (Tang et al., 2020), a radius of 500 m is considered to be a convenient pedestrian area for reaching a seaside. Thus, we defined a coastal area as a waterfront space located within 500 m of a coastline. Over the past two decades, the abrupt urban expansion of Hong Kong's accompanying reclamation schemes has led to a distinct deterioration of local environmental quality and ecosystem services (Cheung and Tang, 2015). To capture the spatial distribution of nature-based activities, from global coastline data provided by the National Geospatial-Intelligence Agency (<https://shoreline.noaa.gov/data/datasheets/pgs.html>), we collected 160,629 geotagged photos and their related metadata from between 2004 and 2020.

To apply a constant metric of photographic intensity (Spalding et al., 2017), we imported the photo-user-days (PUD) function to prevent systematic errors resulting from the same photographer taking multiple photos associated with the same landscape on a single day. Importing the PUD function into the value of CESs better corresponds to Hong Kong's actual annual volume of coastward tourist flows (Sinclair et al., 2018). We then selected 34,855 PUD photographs of the study area. For the analysis of the spatial distribution of CESs, we relied on a PUD density with a grid cell size of 100 m as implemented in ArcGIS 10.4. We calculated PUD densities by dividing the PUD value by the area of the grids.

2.4.3. Content analysis of PUD photos of coastal areas

A content analysis was performed to measure people's preferences for biodiversity and ecosystem services from images posted online (Tenkanen et al., 2017; Hausmann et al., 2018). Using automated object detection tools, scholars have revealed recreational patterns from generated visual information on photos of the natural environment (Grzyb et al., 2021). Specifically, off-the-shelf technologies support semantic image segmentation given their ability to extract latent features from large volumes of information while limiting subjectivity and effort (Zhang et al., 2020).

There were three steps in content analysis of evaluating CESs strength. First, we adopted the Google Cloud Vision API (<https://cloud.google.com/vision/docs/labels>) to detect and indicate dominant entities (e.g., a bicycle, a watercraft, or a bird) in the form of keyword labels from PUD photos (Taecharunroj and Mathayomchan, 2020). A maximum of ten labels were provided for each image, and each label was accompanied by a confidence score. We limited our analysis to a focus on labels with a score of 0.6 or higher (Runge et al., 2020). Second, considering the definition of CESs noted in the above section, we further manually categorized labels associated with 5 or more photos into six categories. Photographs with no natural components related to six broad CESs were excluded from the analyses (e.g., the label "bicycle" represented recreational services, the label "hill station" represented amenity services, and the label "bird" represented naturalistic services). We used 424 labels related to CESs to filter the images, resulting in 12,732 PUD photos uploaded by 3599 users related to CESs. Table A1 in Appendix shows the summary of CES-related keyword labels. The API returned at least one label where a photograph could represent multiple CES categories.

Third, the relative strength of CESs was analyzed following the method proposed by Razis et al. (2020). A dominance value based on the confidence score and the frequency of labels for each image was used. The frequency of label use was calculated according to a logarithmic scale set to prevent values of equal to zero. The mean dominance value

of each CES division was also aggregated over each grid cell. The dominance metric is as follows:

$$\text{Dominance Value} = \sum_{i=1}^n \text{score} \times \lg(1 + n)$$

where $n > 0, n \in \mathbb{Z}$, and n represent the sum of labels related to the specific CES of each photo.

Furthermore, to explore the validity of the CES assessments, five professors of ecology were instructed to classify the components of the photos from a subdataset of 100 randomly selected PUD photos. The experts examined the images for the presence of components related to CES and used the same six categories of CES for further classification. We used Cohen's kappa coefficient (Ruiz-Frau et al., 2020) to assess consistency between the two automated components and examined each CES division individually.

2.4.4. Sentimental analysis of CES-related textual metadata

Sentimental analysis is used to extract information on aesthetic experiences from textual content. Using the machine learning-based approach, researchers have explored how people evaluate or score target entities on online social platforms and have determined the relative correlation between well-being and public green spaces (Do, 2019; Plunz et al., 2019).

Since the text included different languages (English, Mandarin, and Cantonese), we manually translated all textual content into English. We removed trivial and less meaningful text, such as independent numerals, emoticons, and text referring to photographic information (e.g., image numbering or camera types). In particular, semantically meaningless stop words were discarded, such as 'the', 'it', or 'am', and two widely used stop word lexicons were used for further initialization: the Natural Language Toolkit (<https://www.nltk.org/>) and Language Technology Platform (<https://ltp.ai/docs/index.html>).

The sentiments associated with the user-generated textual content are commonly determined by the lexicon-based approach (Su et al., 2021). These lexicons summarize words into positive extremity, neutral, and negative extremity, and also record the expressions for verbs, things, intensifiers, and modifiers. Coupled with machine learning methods, large sets of lexicons are used to train classifiers of sentiment. We thus used the Google Cloud Natural Language API (<https://cloud.google.com/natural-language/docs/analyzing-sentiment>) to determine sentiment associated with the user-generated text (Liu et al., 2020). The sentiment extent was applied to capture general public perception in two areas: the sentiment score and magnitude score. First, the sentiment score was used to quantify the overall emotional learning of the textual content of individuals within each study unit. The sentiment score ranged between -1 (negative) to 1 (positive), and 0 represented neutral. Second, the magnitude score was used to determine how much emotional content (both positive and negative) was present in the crowdsourcing open textual data and showed sentiment strength. The magnitude score was between 0 and positive infinity.

We used the mean sentiment rating to capture an average measure of public perception of ecotourist sentiment and tourism site attractiveness. To overcome potential sampling bias, the mean sentiment and mean magnitude scores of images were aggregated over each grid cell in ArcGIS.

2.4.5. Regression analysis of CESs and related sentiments with demographic and temporal factors

The perceptions of visitors are influenced by their diverse demographic characteristics as well as by light, weather, climate, and other situational factors (Ghermandi and Nunes, 2013; Huang and Wang, 2018). To elucidate the patterns of CES-related photos of coastal areas or nearby locations, we selected various context-specific variables following the related literature. We used three variables from the two categories: (1) two temporal characteristics of tourists' behavior,

including seasonal variations in coastal travel and daytime or nighttime visits for each photo, and (2) the origins of tourists, where Flickr users' origins were separately controlled for local and international tourists.

We obtained detailed results on CESs and related sentiments from the geotagged photos through our regression analysis and multiple comparisons analysis. First, multivariate regression was used to estimate the degrees of significance of different social media characteristics (e.g., the visiting time, season, and user's place of origin) for generating CES potential (Chang Chien et al., 2020; Gosal and Ziv, 2020). Second, the significance of logistic regression was used to determine the presence and extent of the impact of sociodemographic and temporal variables on sentiment ratings (Tang et al., 2021). Third, we tested for statistically significant differences across visitors of various origins and temporal patterns with the Kruskal–Wallis test on ranks (Ghermandi et al., 2020), a nonparametric test that does not require the assumption of a normal distribution. Under the assumption of same shape distributions, we interpreted the result as indicating differences in the medians across groups. When the null hypothesis of the Kruskal–Wallis test was rejected, we used Dunn's post-hoc test with Bonferroni adjustment to draw multiple comparisons as part of a post hoc test for pairwise comparisons across groups. All statistical analyses were carried out in R 4.0.3.

Furthermore, we quantified the frequency of photos with nonzero dominance across all CES-related photos and differentiated based on variables to determine the probability of tourists being associated with specific cultural services. We calculated the mean dominance value of photos by focusing on those with nonzero values only, and then we identified tourists' perceptions of specific cultural services.

3. Results

3.1. Identified CESs through keyword labels

The ten most frequently used labels of each CES class are summarized in Table A1 of the Appendix as well as the other ten labels with the highest scores of each CES division. The main photos show the highest cultural values to pertain to naturalistic and aesthetic services. Almost 70% of all labels focus on coastal landscapes, i.e., photos of coastal environments (entirely or only partly) and associated features.

The labels for cycling (e.g., bicycles, bicycle frames, bicycle wheel rims, and bicycle accessories) and water transportation (e.g., watercraft, boats and boating equipment and supplies, passenger ships, masts, and cruise ships) constitute the topic of recreational services. For aesthetic services, the highest dominance value (1221.91) was found for the label for coastal and oceanic landforms. The labels about resorts (e.g., hill stations, outdoor furniture, trails, field houses, and resort towns) show close relations to amenity services. Most labels related to knowledge services include pollinators (0.29%), maps (0.04%), agriculture (0.03%), bioluminescence (0.02%), and knowledge services pertain to coast popular science. High-scoring photographs of naturalistic values present characteristics of coastal biology. Labels related to naturalistic services often refer to wildlife (e.g., vertebrates, carnivores, invertebrates, or arthropods), plants, and animals as well as to abiotic components (water, land, and stones). Labels associated with historical sites (e.g., sculptures, temples, ruins, monuments, churches, palaces, and graves) form the class of spiritual services.

Table 1 shows the accuracy of the automated detection of CES relative to the manual assessment. For the test subset of 100 random PUD photos, we found a 93% overall classification agreement for

detecting CES (Cohen's kappa = 0.851). The reliability rates exceed the suggested Cohen's kappa coefficient value (0.6) for all six specific case studies, showing fair to a substantial agreement between the two methods. This result implies that the process of automatically identifying the content of geotagged photos by cloud services and of inferring how people perceive CESs proved reliable and statistically significant.

3.2. Spatial heterogeneity of coastal CESs and related sentiments

Fig. 3 shows the spatial distribution of coastal CESs with the photo density. CES-related PUD photos of density in coastal areas of Hong Kong were found to be unevenly distributed. A total of 5.75% of the 753.87 km² grids contain at least one image. From a heterogeneity perspective, hotspots of CESs are primarily concentrated in waterfront areas of Victoria Harbor, Aberdeen Harbor, Tai O, Cheung Chau, Stanley, and Clear Water Bay. The most popular place to take and post cultural photos is Victoria Harbor, accounting for 53.46% of all grids with at least 10 images. Victoria Harbor is a major tourist attraction in Hong Kong with promenades seen as gathering places for both residents and tourists.

Fig. 4(a–f) shows the distribution of dominant CESs in the coastal areas of Hong Kong. Such mapping accounts for the spatial heterogeneity of CESs. Major areas for recreational services are concentrated in several regions, including Victoria Harbor, the Disneyland Resort, and Tolo Harbor, which appear to be of great aesthetic significance to visitors. A grid for Tolo Harbor presents the highest mean dominance value (8.32) of recreational services. Specific archaeological sites and religious architectural areas of Victoria Harbor, Repulse Bay, Tai O, and Cheung Chau host visitors who celebrate the fusion of diverse religious beliefs and practices in a modern city. Photographs associated with amenities and knowledge are sporadic for some areas. In contrast, major areas of naturalistic services are widespread across areas with significant levels of biodiversity, such as the Po Toi Islands, Clear Water Bay, and Sai Kung.

Fig. 5(a and b) shows the spatial distribution of the mean sentiment rating of CES-related photos. Over 60% of the CES-related photographs and textual content show positive and neutral views of coastal areas in Hong Kong.

3.3. Influential factors of CESs and CES-related sentiments

Brief descriptive statistics are performed on the metadata of different subdatasets of PUD photos. Table 2 summarizes the temporal and sociodemographic variables of original unlabeled PUD photos of coastal areas and CES-related photos. Table 2 also shows the descriptive characteristics of the sentiment rating.

A comparison of CES-related photos to unlabeled PUD photos reveals similarities between different groups of tourists (Table 2). We found that tourists had uploaded more photos at night than in the daytime. Interestingly, tourists are more likely to be associated with photographs of CESs from the summer and autumn than those from any other season. Local residents uploaded more PUD photos than international and domestic tourists. CES-related photos taken by locals, to an extent, are less common than original unlabeled photos for all visitors. International tourists from Europe and North America and domestic tourists were found to be particularly active with a relatively similar probability of being associated with elements of CES-related photos.

Table 3 shows the results of the multivariate regression analysis.

Table 1

Cohen's Kappa between manual and automated assessment of CES.

	CES	Recreational services	Aesthetic services	Amenity services	Knowledge services	Naturalistic services	Spiritual services
Accuracy	93%	88%	83%	91%	92%	87%	91%
Value of Kappa	0.851	0.774	0.685	0.816	0.833	0.757	0.816
P value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

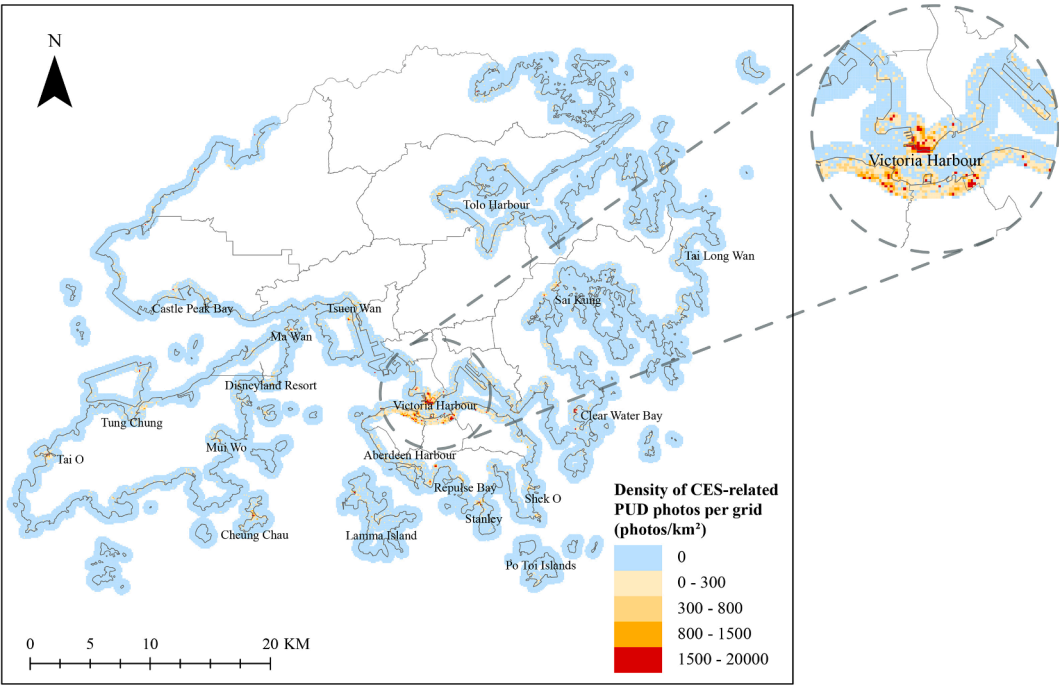


Fig. 3. Distribution of the density of photos related to coastal CESs.

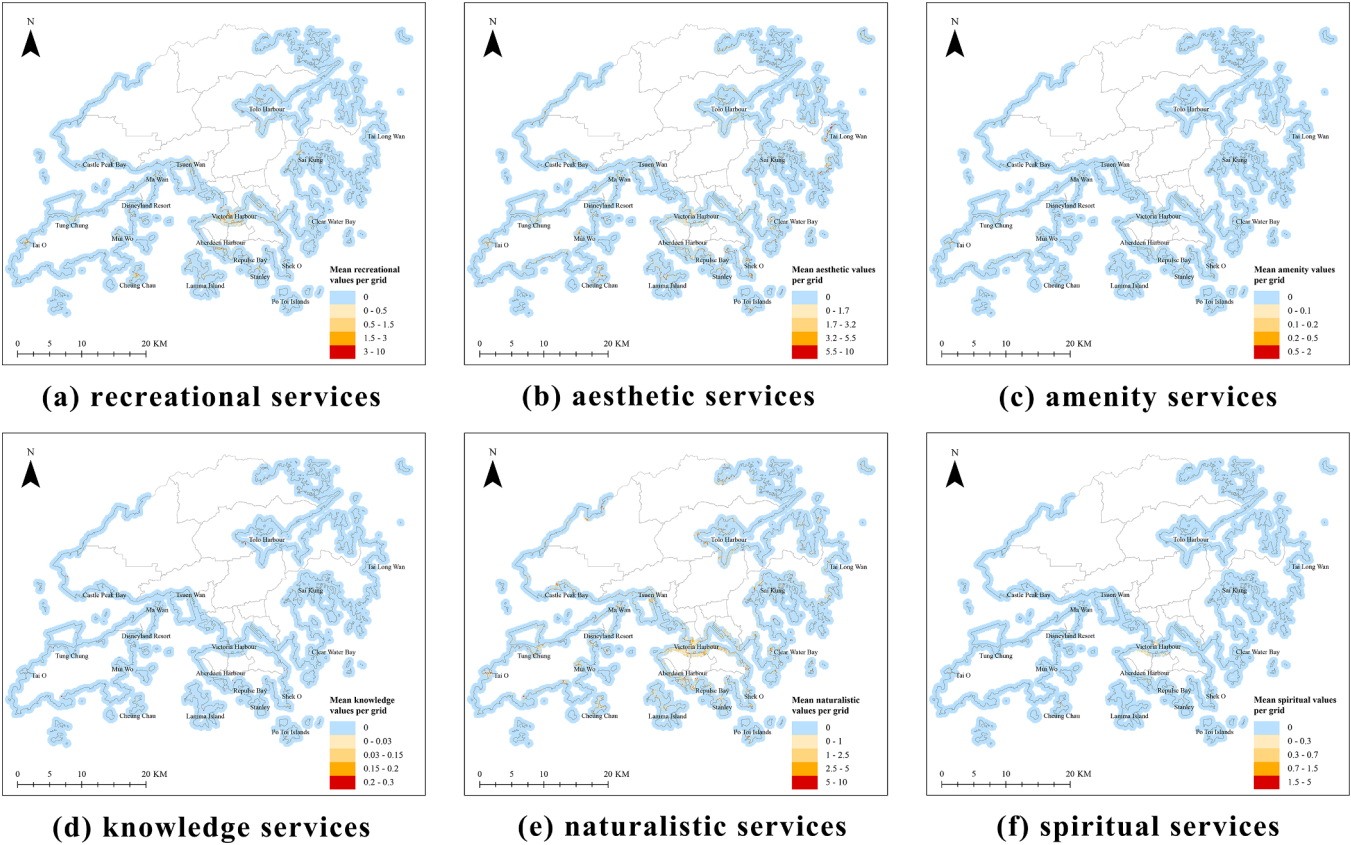


Fig. 4. (a)~(f). Distribution of CESs in the coastal area of Hong Kong.

Tables A2–A4 in the Appendix show CES values differentiated based on temporal and demographic variables from Kruskal–Wallis and Dunn’s tests. We found spring travel to be a statistically positive temporal predictor in the model evaluating spiritual services in contrast to

aesthetic services. Spring travel and daytime visits are more significant positive predictors of naturalistic services than other temporal indicators. The regression analysis and Dunn’s test indicate that recreational services have statistically more positive effects on other groups of

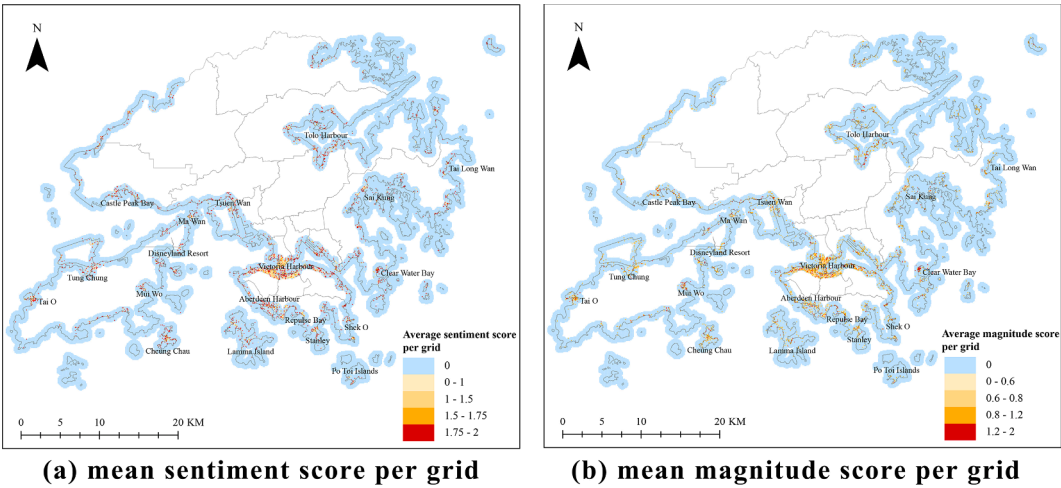


Fig. 5. (a)~(b) Distribution of mean sentiment rating of CES-related photos. Figure (a) and Figure (b) represent the mean sentiment score and the mean magnitude score derived from the textual content of Flickr.

Table 2
Descriptive statistics of social media characteristics.

Variable	Percentage (%) – Original unlabeled PUD photos	Percentage (%) – CES-related photos
Sentiment characteristics		
Sentiment score:	(N = 30,150)	(N = 10,893)
0 = a negative document, (sentiment score <0)	14.0%	13.7%
1 = a neutral document, (sentiment score = 0)	21.9%	22.4%
2 = a positive document, (sentiment score >0)	64.1%	63.9%
Magnitude score:		
0 = no emotional content, (magnitude score = 0)	21.5%	20.4%
1 = mild emotional content, (0< magnitude score <= 0.4),	64.4%	67.8%
2 = strong emotional content, (magnitude score >0.4)	14.1%	11.8%
Travel characteristics		
Day/Night:	(N = 34,855)	(N = 12,732)
0 = daytime (from 6AM to 6PM)	48.8%	47.7%
1 = nighttime (from 6PM to 6AM)	51.2%	52.3%
Season:	(N = 34,855)	(N = 12,732)
0 = spring (from March to May)	25.5%	24.6%
1 = summer (from June to August)	27.1%	25.7%
2 = autumn (from September to November)	24.9%	26.2%
3 = winter (from December to February)	22.5%	23.5%
Hometown:	(N = 19,798)	(N = 6899)
0 = Africa	0.4%	0.4%
1 = Asia except Hong Kong	11.0%	13.5%
2 = Europe	12.5%	13.4%
3 = Hong Kong (Local)	61.3%	57.3%
4 = North America	10.7%	11.5%
5 = Oceania	3.5%	3.4%
6 = South America	0.5%	0.6%

Flickr users than on locals, indicating that local inhabitants are less likely to be associated with photographs of outdoor activities occurring in coastal areas. However, locals are significantly positive predictors of evaluations of naturalistic services.

Table 4 presents the results of the logistic regression analysis. From the logistic regression models for models stratified by daytime and nighttime visits, we found nighttime visits to be a significantly positive predictor of positive and mild sentiment. Visitors from different origins show diverse attitudes on coastal CESs. However, international and domestic visitors, except for Europeans, are statistically more positive predictors of positive sentiment than locals. This result indicates that locals negatively affect where we find mild sentiment magnitudes of Flickr photos in coastal areas.

4. Discussion

4.1. CES assessment result

For the characteristics of coastal CESs, the six categories examined present both similar and different characteristics. With respect to similarities, high concentrations of CESs tend to be more spatially concentrated around both cultural attractions and protected areas. This is attributed to areas with good spatiotemporal accessibility, high infrastructure coverage, and spatially explicit population and economic growth that increase the maximum efficiency of CES extraction or utilization (Ghermandi and Nunes, 2013; Teoh et al., 2019). Such a finding supports the argument that Hong Kong is renowned for its cultural sites (Teoh et al., 2019). Victoria Harbor is the most important tourism destination with rich cultural resources. Photos taken in proximity to Victoria Harbor depict its modern cityscape of water transportation and coastal landscapes.

With respect to differences, for example, there appears to be some correspondence between the locations represented in photographs of high recreational value and cycling path networks in the regions. This observation is consistent with the fact that bike riding is an important outdoor activity among Flickr users (Grzyb et al., 2021). Through the unique strategic marketing decisions made in the global tourism marketplace, Hong Kong distinguishes itself as a destination where a sense of multiculturalism has become a subliminal coopting agent of tourism promotion (Zhang et al., 2015). However, our findings indicate that locals are less likely to be associated with photographs of spiritual services provided in coastal areas. Some scholars have argued that the Hong Kong government did not play an active role in promoting local religious activities in earlier years (Liu, 2003) and that some cultural sites, such as Victoria Harbor, have been less oriented toward locals over the years and have turned to commercial development (Cheung and Tang, 2015).

Table 3

Results of multivariate regression for CES association with temporal variables and Flickr users' origins.

	Recreational services	Amenity services	Aesthetic services	Knowledge services	Naturalistic services	Spiritual services
Seasonal indicators (N = 12,732)						
Summer	0.016	0.0004	0.066*	0.001	−0.263***	−0.025***
Autumn	−0.020	−0.001	0.035	0.0008	−0.223***	−0.009
Winter	−0.018	−0.0003	0.112***	−0.0008	−0.111**	−0.0004
Temporal indicators (N = 12,732)						
Nighttime	0.007	0.003**	0.054*	−0.0008	−0.193***	0.005
Users' origins (N = 6899)						
Africa	0.435**	−0.007	−0.442	−0.006	−0.735**	−0.013
Europe	0.195***	0.002	−0.152***	−0.005***	−0.656***	0.057***
South America	0.207	−0.002	−0.429	−0.006	−0.562*	0.005
Africa	0.435**	−0.007	−0.442	−0.006	−0.735**	−0.013
Europe	0.195***	0.002	−0.152***	−0.005***	−0.656***	0.057***
Oceania	0.286***	0.003	−0.263**	−0.003	−0.604***	0.059***

Notes: ***, ** and * indicate respectively 1%, 5% and 10% statistical significance levels (p-value). "Spring", "Daytime", and "Local" are the omitted variables for the regression model.

Table 4

Results of logistic regression for sentiment rating association with CES based on temporal variables and Flickr users' origins.

	Sentiment Score			Magnitude Score		
	Negative	Neutral	Positive	No emotional	Mild emotional	Strong emotional
Seasonal indicators (N = 10,893)						
Summer	−0.036	0.0001	0.018	−0.0098	−0.071	0.149*
Autumn	0.019	0.0299	−0.032	0.034	−0.107*	0.156**
Winter	0.003	−0.078	0.056	−0.086	0.041	0.045
Temporal indicators (N = 10,893)						
Nighttime	−0.179***	−0.126***	0.187***	−0.09993**	0.248***	−0.330***
Users' origins (N = 6047)						
Africa	0.154	−1.439	1.176	−1.429	0.858	0.572
Asia	−0.557	−2.188**	2.302**	−2.460**	3.360***	−1.702*
Europe	−0.384	1.189	−0.771	1.109	1.244	−3.089***
North America	0.718	−0.593	0.001	−0.697	2.237**	−2.259**
South America	−0.825	−0.379	0.923	−0.367	0.990	−0.916
Oceania	−2.050**	0.136	1.345	0.175	0.505	−0.905

Notes: ***, ** and * indicate respectively 1%, 5% and 10% statistical significance levels (p-value). "Spring", "Daytime", and "Local" are the omitted variables for the regression model.

4.2. CES-related sentiment assessment result

Our results from the analysis of visiting sentiments illustrated on social media provide a comprehensive account of the experiences of various parts of the population. The strength of sentiments associated with CES-related images also differs from the social media characteristics of Flickr. We found that night-time travel may be driving positive sentiment, which is consistent with the fact that Hong Kong is famous for its skyscrapers and the vibrance of its modern cityscape at night (Taecharungroj and Mathayomchan, 2020; Jiang et al., 2021). However, the seasonal variable is not significant. This suggests that seasonality might not be a significant determinant of coastal ecotourism in Hong Kong.

Coastal resorts in Hong Kong are widely attended to get away from crowded cities for tranquility and are directedly linked to a relaxing atmosphere (Liu et al., 2020). Urbanites who post on Flickr in coastal areas usually post positive content and express positive emotions, whether they live in coastal areas or travel to them for fun. The results of this study support observations consistent with previous studies on ecotourism and coastal recreation tourism in Hong Kong (To, 2007). Over the last decade, Hong Kong's ecofriendly coastal conservation policies have contributed significantly to efficient land-use changes in these areas and improved understanding of CESs. Stunning coastal scenes with various natural habitats are particularly popular in Hong

Kong and promote active recreation and water sports culture.

4.3. Implications for coastal ecosystem management

Several implications for sustainable coastal management from our main findings can be summarized. People's demand for coastal amenities should be integrated into the policy discussion and into decision-making processes to yield better-informed strategies for the management of natural capital. Possible solutions include providing more green infrastructure near beaches or conservation areas. Meanwhile, governments should recognize the effects of a vibrant, accessible, and sustainable waterfront to provide seaside space for all residents and tourists.

4.4. Limitations and prospects for future research

We must also acknowledge several limitations of this study. First, in evaluating the results of this study, one must recognize that some subjective judgment is unavoidable in the development of keywords and associations with specific CESs. Notably, according to experts, the viewshed of some photos could complicate the identification of aesthetic and naturalistic services, which also interferes with automated detection software. People usually have a broader field of view when walking

down a coast and thus can enjoy both a broad ocean view and waves around them. Compared to the manual classification approach of CES as well as related sentiments (Alieva et al., 2021), the automated procedure is a compromise between precision and the costs of processing big social media data. Accordingly, future research should focus on the combination of manual analysis of images and content analysis by machine learning tools for a more nuanced examination of the natural environment than a direct description of social media characteristics. Second, geotagged photographs uploaded to social media are uneven in distribution and limited in quantity compared to the huge tourist flows of Hong Kong. Third, digital divides of large-scale studies interfere with research on where ecosystem services actually originate and who is interviewed on social media (Toivonen et al., 2019; Teles da Mota and Pickering, 2020). Therefore, we plan to address social stratification and age demographics in subsequent studies to enrich the composition of samples of social media users.

5. Conclusion

With an increasing demand for CESs, especially in coastal and island cities, there is a need for the systematic and integrated evaluation of ecotourism satisfaction to ensure sustainable planning strategies and the maintenance of ecosystem services. Understanding the sentiment regarding ecotourism is necessary to adequately and equitably plan for long-term coastal ecosystem exploitation in a sustainable manner. The accumulation of geotagged social media data offers new insights into the patterns of CESs from qualitative content analysis. Our paper takes advantage of pluralistic approaches that combine mapping and statistics to fully analyze differences in how CESs are delivered to diverse groups of visitors drawn from social media activities.

The distribution of CESs shows significant spatial heterogeneity, that both cultural attractions and protected areas are typical hotspots associated with CESs, and the assessment of CES-related sentiment demonstrates positive emotions attached to coastal human behaviors. Additionally, the results prove the strong associations among different CESs and related sentiments with sociodemographic and temporal factors. Based on our study, visitors of different origins differ in their cultural preferences of coastal ecosystems as well as the spatial and temporal characteristics of their visitation patterns. Specifically, the present study supports the notion that night-time travel may be driving positive sentiment; however, the seasonal variable is not significant. Overall, the methodological flow of this research is feasible and applicable to other places. In particular, visitors' experiential constructs of the sentimental aspect are demonstrated by our findings. Sentiment evaluation of user-generated text provides insights into users' mental processes, personal conceptualizations, and memories of scenes that cannot be obtained through a sole analysis of an image.

CRedit authorship contribution statement

Haojie Cao: Methodology, Data curation, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Miao Wang:** Supervision, Methodology, Resources, Writing – review & editing. **Shiliang Su:** Conceptualization, Supervision, Methodology, Resources, Writing – review & editing. **Mengjun Kang:** Supervision, Methodology, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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